However, many refusals also arise from prompts that are not flagged as unsafe by the Moderation API—such as dark humor, hypothetical scenarios, or cheating in video games. By comparing user reactions to flagged versus unflagged prompts, we can test whether users accept refusals more readily when the underlying request is clearly inappropriate or unsafe.

4. Results

4.1. LLM-Response-Refusal and Win Rates

Table 4 presents the distribution of LLM response behaviors alongside their corresponding win, loss, and tie rates based on user decisions. We observe a clear refusal penalty, supporting H1, which posits that refusals reduce user satisfaction. Normal responses, which neither refuse nor disclaim, achieve a win rate of 36%, with the remaining outcomes split between losses (33%) and ties (31%). In contrast, responses involving refusals perform significantly worse, indicating a strong user preference against refusal behaviors. While we observe a clear refusal penalty across both ethical and technical refusals, the effect is markedly stronger for ethical refusals—consistent with H2a. Refusal Ethical responses achieve a win rate of only 8%, compared to 16% for Refusal Technical. Similarly, the win/loss ratio for ethical refusals is just 0.16, less than half of that for technical refusals (0.35).

Table 4: LLM Response Moderation and Win/Loss/Tie Rates

	Distribution		User Decision			
Response	#	Share	Win	Loss	Tie	Win/ Loss
Normal	87054	87.16	0.36	0.33	0.31	1.09
Disclaimer Ethical	1235	1.24	0.38	0.34	0.28	1.12
Disclaimer Technical	5730	5.74	0.30	0.39	0.31	0.77
Refusal Ethical	2655	2.66	0.08	0.51	0.41	0.16
Refusal Technical	3202	3.21	0.16	0.46	0.39	0.35

Figure 2 further illustrates conditional win and loss rates based on the response moderation behavior of the opponent model. The distaste for refusal behavior becomes even more pronounced when compared directly to Normal responses: models that refused to answer for ethical reasons have a win rate of only 4% when paired with a model providing a normal response. This finding underscores the strong user preference for models that attempt to answer, even in ethically sensitive contexts. For disclaimed responses, the results are more nuanced. For user decisions, responses labeled as Disclaimer Ethical achieve a win rate of 38% even surpassing the win rate for normal.

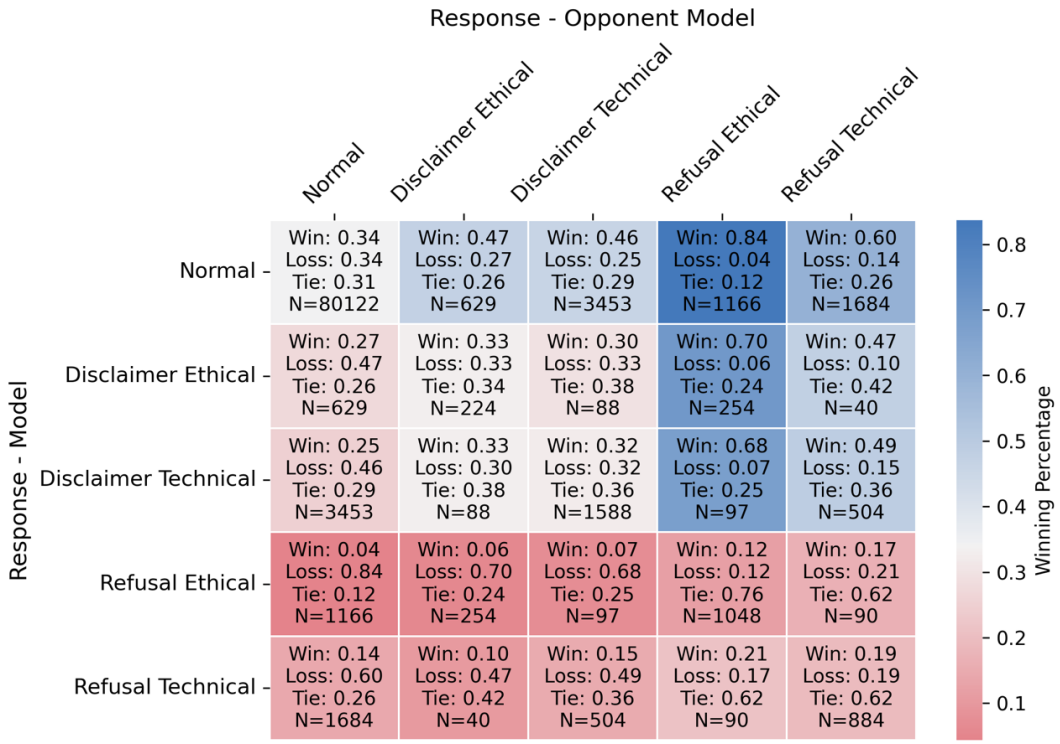

Figure 2. Conditional Win/Loss/Tie Rates by User Decision

However, this high win rate may be partially attributed to pairings with models that outright refuse to answer. When disclaimer responses are compared directly to Normal responses (as shown in Figure 2), the win rates are notably lower with 27% for Disclaimer Ethical. Similarly, for Disclaimer Technical the win rate is overall 30% but drops to 25% in pairings with normal responses, reinforcing that users tend to favor Normal responses over disclaimers or refusals. However, the effect of disclaimers remains clearly weaker than that of full refusals—suggesting that users are more accepting of boundary-setting behavior when it is framed as a qualified attempt to assist, rather than as an outright rejection.

Finally, we estimate the effect of refusal and disclaimer behaviors on win/loss/tie outcomes using OLS regressions, as reported in Table 5. While other modeling approaches, such as logistic regression, are viable, we use OLS for its direct interpretability of coefficients. All regressions control for the LLM model, opponent model, and additional response characteristics, including text length and textual similarity of both responses. For formatting reasons, Table 5 displays coefficients for the focal model only; opponent model coefficients are reported in Appendix Table AT1.

The results show a consistent pattern lending strong support to H1. All refusal (and disclaimer) behaviors are associated with significantly lower win rates and higher loss rates, with stronger effects for refusals compared to disclaimers.

Table 5: OLS Regressions – LLM-Refusals and Win-Rates

	User Decision		
Variable	**Win**	**Loss**	**Tie**
Disclaimer Ethical	-0.073*** (0.019)	0.113*** (0.018)	-0.040*** (0.019)
Disclaimer Technical	-0.098*** (0.009)	0.106*** (0.009)	-0.008 (0.009)
Refusal Ethical	-0.375*** (0.011)	0.320*** (0.013)	0.005*** (0.014)
Refusal Technical	-0.214*** (0.011)	0.162*** (0.012)	0.052*** (0.013)
Similarity	0.006** (0.003)	-0.011*** (0.003)	0.005* (0.003)
Length	0.079*** (0.003)	-0.058*** (0.003)	-0.023*** (0.003)
Controls LLM Model	Yes	Yes	Yes
Controls Opp. Model	Yes	Yes	Yes
Observations	49938	49938	49938
R-squared	0.13	0.13	0.01

Robust standard errors in parenthesis.* p<.1, ** p<.05, ***p<.01. Controls LLM includes dummy variables for the LLM used for the response. Controls Opp. Model represent controls for all listed variables for the opponent model.

In line with H2a, ethical refusals exhibit the largest penalties in user decisions: a refusal for ethical reasons corresponds to a 37 percentage point decrease in win rates and a 32 percentage point increase in loss rates. By comparison, technical refusals lead to a smaller 21 percentage point drop in win rates and a 16 percentage point rise in losses. Wald tests (Wald, 1943; Wooldridge, 2010) confirm that these differences between ethical and technical refusals are statistically significant for both win and loss coefficients ($p < 0.001$).

4.2.Framing and Phrasing of Refusals

To test whether the presentation of refusal responses affects user preferences, Table 6 columns 1-3 report the results of OLS regressions restricted to Refusal Ethical responses. We focus on two linguistic characteristics: text length and textual similarity to the prompt. These serve as proxies for whether the model issues a generic rejection or instead contextualizes the refusal in a more tailored, conversational manner.

The results show that both similarity and length are positively associated with win rates and negatively with loss rates, supporting H3a and H3b. For user decisions, a one standard deviation increase in textual similarity leads to a 1.6 percentage point increase in win rates, while a similar increase in text length results in a substantial 9 percentage point gain.